

Suprakash Biswas

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EDUCATION

Indian Institute of Information Technology, Kalyani
B.Tech in Computer Science and Engineering

Nov 2022 – Present

EXPERIENCE

Software Development Engineer – 5D VDC LLP

Apr 2026 – Present

Reference Link: Offer letter

- Led the end-to-end development of an enterprise analytics platform using React and Node.js, enabling the processing and visualization of over 2 million rows of real-time business data.
- Constructed high-performance rendering modules with TypeScript and Web Workers, offloading heavy data aggregations and decreasing UI render latency by 45%.
- Architected event-driven backend services capable of handling 5,000+ concurrent data ingestion requests while maintaining 99.9% system uptime.
- Optimized database execution plans and query indexing strategies, cutting average query response times by 60% for high-volume data transformation workloads.
- Built custom dashboard state caching layers utilizing Redis and service worker background syncing, minimizing repeated network overhead for static operational views by 35%.
- Implemented automated visual regression testing frameworks alongside Jest, securing 92% component test coverage across dynamic plotting viewports.

Full Stack & ML Intern – Lumineth Private Limited

Apr 2025 – Sep 2025

Reference Link: Internship Certificate

- Shipped 5 core production features for the Auraasync e-commerce platform using React.js and REST APIs, supporting an active user base of 10,000+.
- Converted static Figma designs into responsive UI modules, elevating mobile Lighthouse performance scores from 72 to 96.
- Maintained CI/CD deployment pipelines for a high-traffic live application, reducing average release deployment times by 30%.
- Developed a collaborative filtering-based product recommendation algorithm, deploying via FastAPI to drive an 11% increase in cross-selling conversion rates.
- Integrated automated secure Stripe processing flows and item availability webhooks, resolving multi-tenant inventory deadlocks during peak flash sales.
- Website currently live at :- <https://auraasync.in/>

Full Stack Developer Intern – ByteForge Labs Pvt. Ltd.

Jan 2025 – Apr 2025

Reference Link: Internship Certificate

- Deployed 3 full-stack MERN web applications, accelerating the product launch timeline by 2 weeks in an agile startup environment.
- Formulated 20+ RESTful APIs and modular frontend components, improving codebase maintainability and reducing API payload sizes by 15%.

PROJECTS

Analytical Intelligence Engine – BI Analytics Platform

React, TypeScript, Node.js, Web Workers

- Engineered a full-stack BI application capable of generating interactive dashboards from client datasets exceeding 1 million rows.
- Implemented a Web Worker-based frontend architecture and advanced React state-management, completely eliminating main-thread freezing during heavy computations.
- Programmed backend Node.js aggregation pipelines to process and serve analytical querying workflows in under 800 milliseconds.
- Created dynamic mathematical multi-axis parsing modules to handle unstructured CSV payloads, providing auto-generated schemas for user-facing visualizations.
- Stabilized WebGL canvas graphs to support smooth 60 FPS plotting configurations when rendering real-time high-density scatter charts.

Unified Data Transformation Pipeline (Master Database)

Node.js, PostgreSQL, MongoDB, ETL

- Created a high-throughput ETL pipeline to ingest and consolidate data from 4 heterogeneous sources into a centralized Master Database.
- Automated schema-mapping and normalization algorithms, reducing data inconsistencies by 95% across integrated enterprise systems.
- Refined database indexing strategies in PostgreSQL, accelerating data retrieval rates by 3x for downstream machine learning workloads.
- Deployed distributed message passing streams to buffer processing spikes, eliminating data packet drops under maximum network congestion.

SeekPast – LLM Powered Historical Intelligence Platform

React, Node.js, LangChain, RAG

- Developed an AI-driven intelligence platform leveraging Large Language Models and semantic vector search methodologies.
- Integrated Retrieval-Augmented Generation (RAG) pipelines, increasing the factual accuracy of historical data responses by 40% compared to baseline models.
- Constructed custom prompt chunking filters to combat token truncation errors, optimizing memory tracking context during multi-turn analytical interactions.

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C++, HTML, CSS

Frameworks & Tools: React.js, Node.js, Express.js, FastAPI, Django, Git, Docker, Linux, Vercel, Render

AI/ML & Databases: LLMs, LangChain, RAG, TensorFlow, PyTorch, Scikit-learn, MongoDB, PostgreSQL, MySQL, SQLite